



SOUND-04

Class 09 - Science

1. Match the columns and select the correct option from the codes given here. [1]

Column I	Column II
(a) Work done by all the forces	(i) Change in potential energy
(b) Work done by conservative forces	(ii) Change in kinetic energy
(c) Work done by external forces	(iii) Change in mechanical energy
(d) Work done in uniform velocity	(iv) Zero

- a) (a) - (ii), (b) - (i), (c) - (iii), (d) - (iv) b) (a) - (iv), (b) - (ii), (c) - (i), (d) - (iii)
- c) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i) d) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)
2. Potential energy of an object at a height h is $E_p = mgh$. This equation is true only if [1]
- i. The acceleration due to gravity is constant.
- ii. The potential energy is path dependent.
- a) Only (ii) is true. b) Both (i) and (ii) are false.
- c) Only (i) is true. d) Both (i) and (ii) are true.
3. If the force is applied at an angle θ then work done is: [1]
- a) $W = FS \tan \theta$ and $W = \text{work}$ b) $W = FS \theta$ and $S = \text{Distance}$
- c) $W = FS \cos \theta$ and $F = \text{Force}$ d) $W = FS \sin \theta$ and $W = \text{work}$
4. In a collision: [1]
- a) the momentum of the system is constant b) momentum is always transferred
- c) momentum is always lost d) energy is lost always
5. One joule work is said to be done when: [1]
- a) a force of 1 N displaces a body by 1 cm b) a force of 1 dyne displaces a body by 1 m
- c) a force of 1 N displaces a body by 1 m d) a force of 1 dyne displaces a body by 1 cm
6. In an oscillating simple pendulum: [1]
- a) Only energy transformation takes place b) Work is done by the force of gravity
- c) Work is done by the mass d) Work is done by the string
7. The value of one kWh is: [1]

a) 300 joule

b) 3.6×10^6 joule

c) 3.6×10^3 Joule

d) 3.6×10^8 Joule

8. The work done by a force on a body will be positive if the: [1]

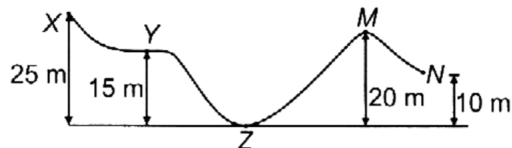
a) body moves perpendicular to the direction of motion

b) body moves opposite to the direction of the applied force

c) the body moves along the direction of the applied force

d) body does not move

9. The diagram shows an object released from point X and travelling on a frictionless track. If a heavier object is taken in consideration then [1]



i. It will move with a greater speed at Y.

ii. The speed of the object at M is 10 m s^{-1} .

a) Neither (i) nor (ii) is true.

b) Only (i) is true.

c) Both (i) and (ii) are true.

d) Only (ii) is true.

10. An electric bulb of 60 W burns for 5 hours a day. The cost of electricity involved in a month of 30 days at ₹ 3.00 per unit is (in ₹) [1]

a) 2.70

b) 2700

c) 270

d) 27

11. A truck and a car having equal kinetic energies are stopped by applying equal retarding force. If S_1 and S_2 are the distances covered by truck and car respectively, before stopping, then [1]

a) $S_1 < S_2$

b) $S_1 = S_2$

c) Cannot say

d) $S_1 > S_2$

12. A tank contains water of mass 'm' to a height 'h'. The potential energy associated is [1]

a) Zero

b) mgh

c) $mg \frac{h}{2}$

d) Depends on the radius of the tank

13. Which of the following quantities have the same units? [1]

a) Power and energy

b) Work and force

c) Power and work

d) Work and energy

14. A skier of mass 50 kg stands at point P at the top of the ski jump and moves from P to Q and takes off his jump at Q as shown in figure. If 50% of the change in the gravitational potential energy of the skier between points P [1]

The diagram shows a curved path representing the trajectory of a projectile. Point P is at the top left, with a vertical line segment next to it labeled '60 m'. Point Q is at the top right, with a vertical line segment next to it labeled '20 m'. The path starts at P, goes down and then up to Q. An arrow on the downward slope indicates the direction of motion. The horizontal line at the bottom is labeled 'Ground'.

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c) Fv

d) $\frac{V}{F}$

22. A mass of M kg is suspended by a weightless string. The horizontal force that is required to displace it until the string making an angle of 45° with the initial vertical direction is [1]

a) $Mg(\sqrt{2} + 1)$

b) $Mg\sqrt{2}$

c) $Mg(\sqrt{2} - 1)$

d) $\frac{Mg}{\sqrt{2}}$

23. The SI unit of energy is _____. [1]

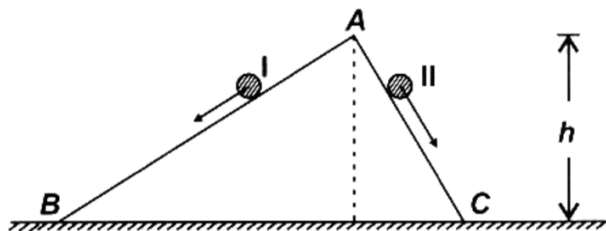
a) joule

b) volume

c) power

d) motion

24. Two inclined frictionless tracks, one gradual and the other steep meet at point A from where two stones (I and II) are allowed to slide down from rest, one on each track as shown in figure. Which of the following statements is correct? [1]



a) Both the stones reach the bottom at the same time but not with the same speed.

b) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.

c) Both the stones reach the bottom at different times and with different speeds.

d) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.

25. If a body is stored at a height h then it will possess: [1]

a) Both potential and kinetic energy

b) Potential energy

c) Kinetic energy

d) Work

26. An elevator with load, weighting 500 kg is moving up with a velocity of 0.2 ms^{-1} . The power of the motor in horsepower is: (1 H.P. = 746 watt) [1]

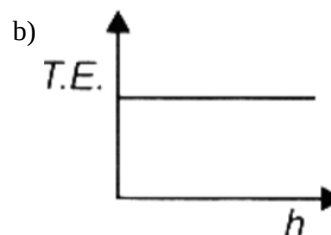
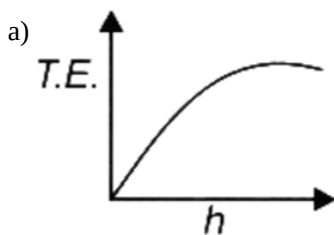
a) 1

b) 1.5

c) 1.3

d) 1.2

27. A graph of the total energy of a freely falling body from a height is [1]



c)

d)

37. In doing the experiment, a student puts the solid iron cuboid in an upright position on loose sand. After that, he has laid it. In these two cases: [1]
- a) the effect of thrust on sand is smaller while in an upright position than lying. b) effects of the thrust are same but different thrusts are there
- c) thrust is the same and its effects are also the same d) thrust is same but effects are different
38. Weight lifter lifts 240 kg from the ground to a height of 2.5 m in 3 seconds. His average power is: [1]
- a) 1960 W b) 196 W
- c) 1.96 MW d) 19.6 kW
39. A body is falling from a height h . After it has fallen a height $\frac{h}{2}$, it will possess [1]
- a) only potential energy b) more kinetic and less potential energy
- c) only kinetic energy d) half potential and half kinetic energy
40. One unit of electricity is consumed by A : P = 40 W bulb used for $t = 25$ hours [1]
 B : P = 20 W bulb used for $t = 50$ hours
 Both A and B are true as energy E in kWh is best related as
- a) $E = Pt$ b) $E = \frac{t}{1000}$
- c) $E = \frac{pt}{1000}$ d) $E = \frac{p}{100}$
41. Which one of the following is not the unit of energy? [1]
- a) joule b) kilowatt hour
- c) kilowatt d) newton metre
42. Which of the following statements is true for the pressure exerted by a solid body? [1]
- a) Greater the area over which the force acts, greater is the pressure b) It does not depend upon the area of contact for the magnitude of force
- c) It depends upon the area of contact for the same magnitude of the force: d) It does not depend upon the area of contact for the same magnitude of force
43. When the speed of a moving object is doubled, its: [1]
- a) kinetic energy increases 4 times b) kinetic energy is doubled
- c) acceleration is doubled d) weight is doubled
44. Two masses m and $2m$ are dropped from a certain height ' h '. Then on reaching the ground, [1]
- a) K.E. of the lighter is 4 times the K.E. of the heavier b) K.E. of the heavier is 4 times the K.E. of the lighter
- c) K.E. of the heavier is more than that of the lighter. d) K.E. of them will be equal
45. A girl is carrying a school bag of 3 kg mass on her back and moves 200 m on a leveled road. The work done against the gravitational force will be ($g = 10\text{m/s}^2$) [1]
- a) 6 J b) 0.6 J

- c) zero d) $6 \times 10^{-3} \text{ J}$
46. A rickshaw puller pulls the rickshaw by applying a force of 100 N. If the rickshaw moves with a constant velocity of 36 km/h. Find the power of rickshaw puller. [1]
- a) 10 Watt b) 100 Watt
c) 2000 Watt d) 1000w
47. Water stored in a dam possess [1]
- a) potential energy b) no energy
c) electrical energy d) kinetic energy
48. Two discs of the same radii and the same mass are resting over one another co-axially on the sand. If one disc is removed gently, then the pressure exerted by the single-disc will be: [1]
- a) doubled b) remains the same
c) half d) can't be predicted
49. A body of mass 2 kg is thrown up vertically with a kinetic energy of 490 J. If the acceleration due to gravity is 9.8 m s^{-2} , the height at which the kinetic energy of the body becomes half of the original value is [1]
- a) 10 m b) 50 m
c) 12.5 m d) 25 m
50. Coolie A takes 1 min to raise a box of mass m through a height of 2.0 m. Another coolie B takes 30 s for the same job. Which one has greater power? [1]
- a) Both Coolie A and Coolie B b) Coolie B
c) Cannot say d) Coolie A
51. A ball dropped from a height of 10m. if the energy of the ball reduces by 40% after striking the ground, how much high can the bounce back? ($g = 10 \text{ m/s}^2$) [1]
- a) 6 m b) 5 m
c) 0 m d) 3 m
52. A boy of mass 50 kg runs up a staircase of 45 steps in 9 s. If the height of each step is 15 cm, find his power. [1]
Take $g = 10 \text{ ms}^{-2}$.
- a) 400 W b) 560 W
c) 270 W d) 375 W
53. A boy of mass 50 kg runs up a flight of 120 stairs, each measuring a height of 25 cm, in one minute and 40 seconds. If $g = 10 \text{ m s}^{-2}$, the power of the boy is [1]
- a) 250 W b) 750 W
c) 150 W d) 500 W
54. A truck of mass 800 kg generates a power of 20000 W. How much time does the truck need to accelerate from a speed of 20 m s^{-1} to 30 m s^{-1} ? [1]
- a) 10 s b) 7.5 s

c) 5 s

d) 6.3 s

55. When a body is whirled in a circle the work done by the centripetal force is: [1]

a) negative

b) zero

c) positive

d) infinite

56. On tripling the speed of motion of a body, the change in K.E. is: [1]

a) 8 times

b) 2 times

c) 9 times

d) 4 times

57. To observe and compare the pressure exerted by solid iron cuboid on the sand, a student takes a cuboid of 5 kg and has a dimension of its base 30 cm $\times$ 15 cm and measures its pressure, then he mounts the cuboid on a cylinder having a radius of base 7 cm and weight equal to 1 kg. The pressure of the whole set-up on the sand is: [1]

a) decreased by 3.5 times

b) remains same

c) can't say

d) increased by 3.5 times

58. The cgs. unit of work is: [1]

a) watt

b) joule

c) dyne

d) erg

59. The heart does 1.5 J of work in each heartbeat. How many times per minute does it beat if its power is 2 watt? [1]

a) 80 times

b) 60 times

c) 8 times

d) 10 times

60. The potential energy of a person is minimum when: [1]

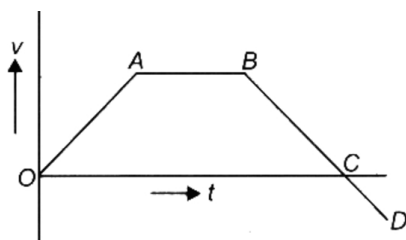
a) a person is sitting in a chair

b) a person is sitting on the ground

c) a person is lying on the ground

d) person is standing

61. A body moves under a variable force. For the body a plot of velocity versus time is shown in figure. The correct statement is [1]



a) In moving from B to C, work done by the force on the body is positive

b) In moving from A to B, the body does work on the system

c) In moving from O to A, work done by the force on the body is negative

d) In moving from C to D, work done by the force on the body is positive

62. An object of mass 1 kg has a potential energy of 1 joule relative to the ground when it is at a height of: [1]

a) 1 m

b) 32 m

c) 9.8 m

d) 0.102 m

63. An iron sphere of mass 30 kg has the same diameter as an aluminum sphere whose mass is 10.5 kg. The spheres are dropped simultaneously from a cliff. When they are 10 m from the ground, they have the same. [1]

- a) kinetic energy
- b) potential energy
- c) acceleration
- d) momentum

64. **Statement 1:** A man carrying a bucket of water and walking on a rough level road with a uniform velocity does no work while carrying the bucket. [1]

Statement 2: The work done on a body by a force (F) in giving it a displacement (S) is defined as $W = \vec{F} \cdot \vec{S} = FS \cos \theta$, where θ is the angle between vectors $\vec{F}$ and $\vec{S}$.

- a) Statement 1 is true but statement 2 is false.
- b) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.
- c) Both statements 1 and 2 are false.
- d) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

65. A mass m is moving in a circle of radius 1 m with a uniform speed of 5 m/s. The work done when its radius turns by 60° is: [1]

- a) zero
- b) 10 joule
- c) 2.5 joule
- d) 5 joule

66. **Statement 1:** If a light body and a heavy body possess the same momentum, the lighter body will possess more kinetic energy. [1]

Statement 2: The kinetic energy of a body varies as the square of its velocity.

- a) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.
- b) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.
- c) Both statements 1 and 2 are false.
- d) Statement 1 is true but statement 2 is false.

67. Calculate the potential energy of a person having 60 kg mass on the summit of Mt. Everest. Height of Mt. Everest is 8848 m from sea level. ($g = 9.8 \text{ m/s}^2$). [1]

- a) 52,02,624 joule
- b) 25 joule
- c) 19.7 joule
- d) 2,02,624 Joule

68. In case of negative work the angle between the force and displacement is: [1]

- a) 180°
- b) 90°
- c) 0°
- d) 45°

69. _____ is the SI unit of work. [1]

- a) joule
- b) energy
- c) work
- d) mass

70. If the force applied on the body distances the body in the opposite direction of force applied then work done is: [1]

- a) Negative
- b) Positive
- c) Constant
- d) Zero

71. A total of 4900 J was spent in lifting a mass of 50 kg. the mass was raised to a height of: [1]

- a) 10 m
- b) 245,000 m
- c) 960 m
- d) 98 m

72. In the explosion of a cracker/bomb: [1]

- a) momentum is conserved
- b) energy is produced
- c) momentum remains unconserved
- d) internal energy is absorbed

73. To observe and compare the pressure exerted by solid iron cuboid, a student marked all the faces of cuboid from 1 to 6. After doing the experiment with all faces of cuboid, he observed that [1]

- a) adjacent faces of cuboid exerted the same pressure
- b) all the faces of cuboid exerted same pressure on the sand
- c) opposite parallel faces of cuboid exerted unequal pressure on the sand
- d) opposite parallel faces of the cuboid exerted some pressure on the sand

74. A microphone converts sound energy to: [1]

- a) A pulse
- b) Sound energy
- c) Electrical signal/energy
- d) A wave

75. The potential energy of a body is by virtue of its _____. [1]

- a) direction
- b) power
- c) position
- d) energy